Abstract Linear Algebra Homework3

intestellartrek

July 1, 2026

Problem1

By observation, for the linear mapping T: $\mathbb{F} \to \mathbb{F}$, the linear mapping are multiplication maps $T_a(x) = ax$. Because $\mathbb{F}$ is actually a one-dimensional vector space and we easily find out the basis is 1

Proof.

Claim. *Every such map is linear*

Let T: $\mathbb{F} \to \mathbb{F}$ be F-linear. Fix $c \in \mathbb{F}$ and define $T(x) = cx$. For $x, y \in \mathbb{F}$ and $b \in \mathbb{F}$,

$$T(x + y) = c(x + y) = cx + cy = T(x) + T(y),$$

$$T(bx) = c(bx) = b(cx) = bT(x).$$

So T is linear.

Claim. *Every linear map has this form*

Let T:$\mathbb{F} \to \mathbb{F}$ be linear, and set c:=$T(1) \in \mathbb{F}$. For any $x \in \mathbb{F}$, since F is one dimensional as a vector space over itself, $x = x \cdot 1$ (scalar x times the vector 1). By linearity,

$$T(x) = T(x \cdot 1) = x \cdot T(1) = xc = cx.$$

Thus T agrees with multiplication by $c = T(1)$ on all of F.

Combining both directions, the F-linear maps $T : F \to F$ are precisely $T(x) = cx$ for $c \in F$, and distinct values of c give distinct maps (since $c = T(1)$ is determined by T). □

Problem2

(a) T is linear

Proof. Let T:$V \to V$ be a function that sends a polynomial to its derivative. For $p, q \in V$ and $c \in \mathbb{R}$

$$T(p + q)(x) = (p + q)'(x) = p'(x) + q'(x) = T(p)(x) + T(q)(x)$$

$$T(\lambda p)(x) = (\lambda p)'(x) = \lambda(p'(x)) = \lambda T(x)$$

such that, T is a linear map. □

(b) T is not injective

Proof. We consider the constant polynomial $p(x) = 1$, such that the function send it to 0 by the derivative. Which means $\ker(T) \neq \{0\}$, actually we can easily find out $\ker(T)$ is the set of all constant polynomial. Such that T is not injective. □

(c) T is surjective

Proof. Since according to the calculas, for polynomials with coefficient is real number, they always has a anti-deriative, so every image has a preimage. Such that T is surjective. □

(d)

The result we prove in the class is actually require the dimension of V is finite. Since $V = \mathbb{R}[x]$, the basis is $\{1, x, x^2, x^3...\}$ which is infinite. So the hypothesis contradiction to the Theorem, and there is no contradiction if the vector space is finite-dimensional that a map is surjective but not injective.

Problem3

Recall the rank-nullity Theorem. $dim(V) = rank(T) + null(T)$

(a)

Proof. If T is injective, that implies us that $\ker(T)$ is actually zero which means

$$dim(V) = rank(T)$$

and since $rank(T) \leq dim(W)$, we have the conclusion $dim(V) \leq dim(W)$ □

(b)

Proof. If T is surjective iff $rank(T) = dim(W)$, such that we have,

$$dim(V) = rank(T) + null(T) = dim(W) + null(T)$$

which implies

$$dim(w) \leq dim(V)$$

□

(c)

Proof. If T is bijective, which implies

$$null(T) = 0, rank(T) = dim(W)$$

combine with the rank-nullity theorem,

$$dim(V) = dim(W)$$

□

Problem4

Let $T : V \to V$ be a linear map with T^2=0

Proof. Let v be an arbitary element $\in im(T)$, by the definition of mapping, there exist $u \in V$ such that $v = T(u)$. Then,

$$T(v) = T(T(u)) = T^2(u) = 0,$$

Thus we have $T(v)$=0, by definition, $v \in ker(T)$

Since v is arbitary we can conculde that $im(T) \subseteq ker(T)$

□

Problem14

Reecall the rank-nulity theorem, we have $dim(V) = rank(T) + null(T)$.

Proof. Assume there is a such a linear map $T : \mathbb{F}^5 \to \mathbb{F}^2$. If the null space is $\{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{F}^5 : x_1 = 3x_2 and x_3 = x_4 = x_5\}$ such that we can rewrite it as $\{(3x_2, x_2, x_3, x_3, x_3) \in \mathbb{F}^5\}$ Thus we can find out the basis for bull space which is $\{(3, 1, 0, 0, 0), (0, 0, 1, 1, 1)\}$. Such that $ker(T)$=2. By the rank-nulity theorem, since $dim(\mathbb{F}^5) = 5$, so the $rank(T)$=5-2=3, since the range is actually a subspace of $\mathbb{F}^2$, such that $rank(T) \leq dim(\mathbb{F}^2)$, thus we have 3 bigger than 2 which is false. Such that there does nit exist a linear map have those null space. □

Check-in

I went to Stanford and find out the taste of cafeteria there is pretty much the same as the Crossroads!